Query ratios · descriptive action endpoint

CPU Polars 2.0.0 / GPU · Single H100 study host · fixed configuration · four engine paths, five software configurations

GPU Spark 3.5.9 · baseline

	SF1	SF10	SF30	SF100	SF300
Q1 · grouped scan	0.13×	0.92×	1.65×	2.08×	2.26×
Q3 · three-table join	0.05×	0.19×	0.38×	0.53×	0.58×
Q6 · filtered sum	0.04×	0.21×	0.40×	0.54×	0.60×
Q10 · four-table join	0.06×	0.12×	0.21×	0.40×	0.49×
Q12 · conditional counts	0.03×	0.11×	0.18×	0.28×	0.42×
Q14 · part join	0.03×	0.12×	0.25×	0.42×	0.61×

GPU Spark 3.5.9 · complex

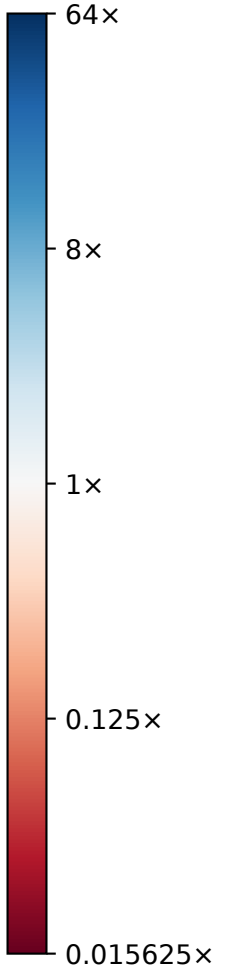
	SF30	SF100	SF300
Q9 · six-table join	0.30×	0.20×	0.15×
Q13 · outer join	0.51×	1.30×	1.83×
Correlated semi/anti	0.26×	0.57×	0.75×
LAG / ROW_NUMBER	1.62×	3.17×	5.15×
Exact distinct	0.49×	0.59×	0.63×
Fact self-join	0.42×	0.69×	0.85×

Polars GPU 1.44.2 · baseline

	SF1	SF10	SF30	SF100	SF300
Q1 · grouped scan	0.78×	2.71×	3.63×	3.94×	3.92×
Q3 · three-table join	0.50×	1.62×	2.73×	3.16×	2.60×
Q6 · filtered sum	0.24×	0.77×	0.95×	1.10×	1.16×
Q10 · four-table join	0.31×	0.94×	1.82×	2.46×	2.56×
Q12 · conditional counts	0.26×	0.69×	1.00×	1.34×	1.94×
Q14 · part join	0.20×	0.64×	0.88×	1.22×	1.51×

Polars GPU 1.44.2 · complex

	SF30	SF100	SF300
Q9 · six-table join	2.73×	3.89×	3.60×
Q13 · outer join	N/A	N/A	N/A
Correlated semi/anti	1.58×	2.38×	2.56×
LAG / ROW_NUMBER	N/A	N/A	N/A
Exact distinct	3.30×	4.40×	4.60×
Fact self-join	1.94×	1.89×	1.52×



Cells show geometric means of ten paired block query ratios for CPU Polars 2.0.0 / GPU, using the action endpoint. Color encodes log2(ratio), centered symmetrically on 1×. These query comparisons are descriptive; no query significance tests or multiplicity-adjusted query claims are made. Gray N/A cells retain the original strict GPU regex and grouped-window rejections. They are not a ratio of 1 and receive no imputed latency. The original correlated query does not establish an executed nonempty anti join.

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